HackInsight 2026 — Executive Summary

PM2.5 Forecasting & Clean Air Decision Support System

Team: Khubsoorat Shklein (Data-Insight)
Track: Air Quality | Release: v1.0-final
User: City Environmental Protection Officer (EPO)

1. Problem Statement & Operational Impact
Severe seasonal smog across Pakistani cities (Lahore, Peshawar, Islamabad, Karachi) poses fatal acute health emergencies. City EPOs previously relied on fragmented static monitors with zero proactive lead-time. We deliver an automated, uncertainty-aware 24h forecasting engine with cited generative intelligence to trigger school closures, traffic rerouting, and industrial anti-smog advisories 24 hours in advance.

The 5 Delivered System Pillars:
1. Warehouse: 3-layer DuckDB (338k PM2.5 + 70k weather records)
2. DQ Agent: Automated fix/fill/flag decision engine with audit log
3. Forecasting Model: Hybrid LightGBM+CatBoost with 84% intervals
4. Insight Agent: RAG grounded in 7 verified PDFs with row IDs
5. Automation: Hourly Prefect pipeline + FastAPI backend + React UI

2. Model Benchmarking, Uncertainty & Regional Fairness (Test Set: 9,443 rows, Jul–Oct 2026)

Model Evaluation	Overall MAE	RMSE	MAPE	84% Interval Coverage	vs. Declared Baseline
Persistence Baseline (y[t] = y[t-24h])	11.65 µg/m³	17.74	34.74%	N/A (Point forecast)	Baseline reference
Hybrid Ensemble (LightGBM + CatBoost)	**2.78 µg/m³**	**4.31**	**9.84%**	**84.10% (calibrated)**	**76.14% Improvement ■**

City Fairness Breakdown: **Karachi:** MAE 1.48 | Cov 84.9% | **Islamabad:** MAE 2.42 | Cov 84.5% | **Lahore:** MAE 3.45 | Cov 83.8% | **Peshawar (Weakest):** MAE 3.77 | Cov 83.2%

3. Autonomous Agents & Responsible AI (Zero Hallucination Guarantee)

Data Quality Agent (src/dq_agent)
• **Action Matrix:** Evaluates schema, timeliness, spikes, and missing values. Applies **FIX** (reversible type/unit fixes), **FILL** (linear interpolation on gaps ≤2h), and **FLAG** (marks spikes >500 µg/m³ & gaps >24h without discarding).
• **Audit Trail:** All actions timestamped with row counts in quality_log.jsonl. 0 unrecorded transformations.

Grounded Insight Agent (RAG) (src/rag & backend)
• **Evidence Corpus:** 247 vector chunks indexed across 7 authoritative sources (WHO AQG 2021, Pak EPA Smog Advisory, PAQI Reports, Health Studies).
• **Row ID Grounding:** Numbers inserted strictly from DuckDB gold rows by code (never LLM generation). Off-topic queries trigger honest *'I don't know'*.
• **Pakistan Impact Bonus:** Dual-language English and Urdu health advisories.

4. End-to-End Automation, Architecture & Business Sustainability

Pipeline & Production Architecture:
• **Hourly Orchestration:** Prefect flow (5 * * * *) fetches OpenAQ v3 & ERA5, cleans, processes, gates, and generates 96 forecasts in ~5.3 mins.
• **Production Serving:** FastAPI REST API deployed on Render with persistent Parquet forecasts and CORS allowlist.
• **Frontend Dashboard:** React 19 + Vite deployed on Vercel at https://data-insight-two.vercel.app/ featuring interactive intervals and chat.

Business Model (Value, Users, Sustainability):
• **Primary User:** Provincial/City EPA Officers and School District Planners.
• **Payer:** Municipal Health & Environmental Authorities (Grant/Contract model).
• **Quantifiable Value:** Estimated **$2.4M annual healthcare savings** in respiratory admissions through 24h preemptive smog alerts.
• **Operating Cost:** ~$18/month (Cloud hosting & open APIs; free tier during prototyping).

Final Verification: ■ make data verified from clean state | ■ Manifest SHA-256 complete (17 files) | ■ Baseline declared early & beaten by 76.14% | ■ 24h Prefect history logged | ■ 13/13 backend unit tests passed | ■ Live on Vercel & Render | **Signed:** Team Khubsoorat Shklein